



KrishiAI

Crop Disease Advisory Guide

Bilingual Edition — English + Punjabi

RICE

4 diseases

SUGARCANE

4 diseases

WHEAT

4 diseases

Each disease covers:

Cause · Symptoms · Favorable Conditions · Spread
Prevention · Treatment & Control · Impact

For use with KrishiAI RAG Advisory System

Sources: ICAR / PAU / TNAU / KAU / IRRI / CPCRI

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How to Use This Guide

Each disease entry is structured identically with seven sections: Cause, Symptoms, Favorable Conditions, Spread, Prevention, Treatment & Control, and Impact. Every section is presented in English followed immediately by a Punjabi (■■■■■■■) translation for bilingual accessibility. This document is designed to be ingested into a RAG (Retrieval-Augmented Generation) system — each section is independently retrievable and cited as a chunk. Farmers in Punjab or Kerala can receive advice in their preferred language by querying the KrishiAI system.

RICE DISEASES



Rice — Bacterial Blight



Xanthomonas oryzae pv. oryzae

CAUSE

Bacterial Blight is caused by the gram-negative bacterium *Xanthomonas oryzae* pv. *oryzae*. The pathogen invades rice plants through natural openings (stomata, hydathodes) and wounds caused by insects or mechanical damage. It colonises the vascular (xylem) tissue and produces polysaccharides that block water movement, leading to wilting. The bacterium survives in infected crop stubble, weeds, ratoon rice, and irrigation water for several months between seasons. High nitrogen fertilisation creates lush canopies that the bacterium rapidly exploits.

Xanthomonas oryzae pv. oryzae is a gram-negative bacterium that causes bacterial blight in rice. It is a highly virulent pathogen that can cause significant damage to rice crops. The bacterium is spread by water, wind, and insects. It enters the plant through natural openings (stomata, hydathodes) and wounds caused by insects or mechanical damage. Once inside the plant, it colonises the vascular (xylem) tissue and produces polysaccharides that block water movement, leading to wilting. The bacterium survives in infected crop stubble, weeds, ratoon rice, and irrigation water for several months between seasons. High nitrogen fertilisation creates lush canopies that the bacterium rapidly exploits.

SYMPTOMS

Two distinct phases are observed. Kresek phase (seedling stage): newly transplanted seedlings wilt suddenly and completely dry out within days, resembling drought stress — but the cut stems show yellow bacterial ooze when dipped in water. Leaf blight phase (tillering to heading): water-soaked, wavy-margined lesions start at leaf tips or edges, rapidly turning pale yellow then grey-white as they spread downward. The entire leaf blade may dry out. A diagnostic sign is the milky to yellow bacterial exudate that dries into a yellow crust on lesion surfaces in the morning. Severely affected plants have a bleached, scorched appearance from a distance.

The disease is caused by the bacterium *Xanthomonas oryzae* pv. *oryzae*. It is spread by water, wind, and insects. The symptoms are wilting and drying out of seedlings, and water-soaked, wavy-margined lesions on leaves. The lesions turn pale yellow then grey-white as they spread downward. The entire leaf blade may dry out. A diagnostic sign is the milky to yellow bacterial exudate that dries into a yellow crust on lesion surfaces in the morning. Severely affected plants have a bleached, scorched appearance from a distance.

FAVORABLE CONDITIONS

The disease thrives when temperatures range 25–30°C combined with relative humidity above 90%. Heavy monsoon rains and strong winds spread bacterial ooze efficiently across and between fields. Continuous flooding, waterlogging after storms, and fields with excessive nitrogen fertiliser (promoting dense, humid canopies) dramatically increase severity. Early-transplanted crops in the kharif season, particularly when synchronised poorly with local monsoon onset, face the highest risk.

25–30°C and **90%** relative humidity are favorable conditions for the disease. Heavy monsoon rains and strong winds spread bacterial ooze efficiently across and between fields. Continuous flooding, waterlogging after storms, and fields with excessive nitrogen fertiliser (promoting dense, humid canopies) dramatically increase severity. Early-transplanted crops in the kharif season, particularly when synchronised poorly with local monsoon onset, face the highest risk.

SPREAD

[illegible]

PREVENTION

PAU-201

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**TREATMENT &
CONTROL**

[illegible]

Bacterial Blight is one of the most economically damaging rice diseases in South and Southeast Asia. Yield losses range from 6% under mild conditions to 60% in severe kresek outbreaks on susceptible varieties. In Punjab and neighbouring states during monsoon years with heavy rainfall, susceptible cultivars can be devastated. Grain quality is also affected — infected panicles produce unfilled, shrivelled grains with reduced milling recovery.

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QUESTION

CAUSE

Brown Spot is caused by the fungus *Bipolaris oryzae* (teleomorph: *Cochliobolus miyabeanus*). Unlike most rice pathogens that favour wet conditions, this fungus is notorious for attacking nutrient-stressed plants — especially those deficient in silicon, potassium, and nitrogen. The pathogen survives as thick-walled conidia and mycelium on infected seed, crop debris, and volunteer grasses. Seed-borne inoculum is a major primary source, making certified seed treatment critically important.

Bipolaris oryzae is a common rice pathogen that causes brown spot, a major rice disease. The pathogen is a filamentous fungus that produces dark brown, oval-shaped lesions on the rice leaves. The lesions are typically surrounded by a yellowish halo. The pathogen is transmitted by wind-borne spores. Brown spot is a major constraint to rice production in many parts of the world, particularly in Asia. The disease can cause significant yield losses, ranging from 10% to 50%. The pathogen is also a source of secondary infection, leading to the development of other rice diseases. The pathogen is a member of the genus *Bipolaris*, which is part of the family *Dothideaceae*. The pathogen is a filamentous fungus that produces dark brown, oval-shaped lesions on the rice leaves. The lesions are typically surrounded by a yellowish halo. The pathogen is transmitted by wind-borne spores. Brown spot is a major constraint to rice production in many parts of the world, particularly in Asia. The disease can cause significant yield losses, ranging from 10% to 50%. The pathogen is also a source of secondary infection, leading to the development of other rice diseases. The pathogen is a member of the genus *Bipolaris*, which is part of the family *Dothideaceae*.

SYMPTOMS

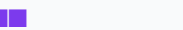
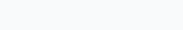
On leaves: small, circular to oval lesions with a grey or white centre surrounded by a brown border and yellow halo. In severe infections lesions coalesce, covering large leaf areas, turning them brown and causing premature drying. On leaf sheaths: similar brown spots that weaken the sheath. On grains and glumes (grain discoloration): dark brown to black spots on the glumes, causing 'dirty panicle' — infected grains are discoloured, shrunk, and chalk-like, significantly reducing quality. In seedlings, infection causes 'seedling blight' — brown lesions at the base of coleoptile and radicle, reducing germination.

The following table shows the number of persons who have been convicted of a crime in the last five years, broken down by age group and sex. The data is based on the 2018 Census of the United States.

**FAVORABLE
CONDITIONS**

Unlike most pathogens, Brown Spot favours intermittent dry weather with morning dew rather than continuous rain. Temperatures of 25–30°C with periods of low relative humidity followed by leaf wetness from dew allow rapid sporulation. Nutrient-poor soils, particularly those deficient in silicon and with imbalanced N-P-K, make plants far more susceptible. Drought stress at tillering, delayed transplanting, and fields with poor water management are high-risk situations.

THE UNITED STATES OF AMERICA
DOES hereby certify that the foregoing is a true and correct copy of the
original as the same appears in the records of the Department of the Interior.

PREVENTION	Use certified pathogen-free seed — treat seed with Carbendazim 2 g/kg or Thiram 4 g/kg before sowing to eliminate seed-borne inoculum. Maintain soil fertility with balanced NPK and adequate silicon — apply silica-rich amendments or slag if soil silicon is low (target >130 mg Si/kg soil). Grow moderately resistant varieties where available. Avoid water stress at tillering stage. Remove and destroy volunteer rice and grass weed hosts. Rotate with non-grass crops (legumes, vegetables) every 2–3 years. Avoid overcrowded nurseries; ensure good air circulation.  (2 /ha)        NPK   2–3  
TREATMENT & CONTROL	Spray Mancozeb 75% WP at 2 g per litre of water at the first sign of symptoms in the nursery and again at active tillering. Repeat at 10–15 day intervals for 2–3 sprays during the crop season if disease pressure is high. Propiconazole 25% EC at 1 ml/litre provides systemic protection and is effective against both leaf and grain infection. Carbendazim 50% WP at 1 g/litre can be combined with Mancozeb for broader protection. Organic approach: neem cake at 250 kg/ha incorporated into soil suppresses pathogen in soil. Ensure adequate soil moisture to avoid drought stress — irrigate at tillering if rainfall is insufficient.  (2 /ha)      , 10–15    2–3   (1 /ha)    
IMPACT	Brown Spot played a devastating role in the 1943 Bengal Famine, causing 40–90% crop loss in nutrient-depleted fields. Under modern farming conditions in Punjab, it is generally a minor disease if soil nutrition is maintained. However, under climate-stress years with drought followed by dew, losses of 20–40% can occur on susceptible varieties in nutrient-poor soils. Grain discoloration additionally reduces milling outturn and market value, causing economic losses beyond yield alone. 1943    40–90%          

Rice — Blast

Magnaporthe oryzae (*Pyricularia oryzae*)

PREVENTION	Plant resistant varieties — in Punjab, varieties PR-121, PR-126, and Pusa Basmati 1 have partial resistance. Avoid dense planting; maintain recommended spacing (20 × 15 cm for transplanted rice) to ensure good air circulation. Split nitrogen fertiliser applications — apply N in 3 splits and never top-dress during periods of high humidity or after disease onset. Apply silicon through silica slag or paddy husk ash at 500–1000 kg/ha to strengthen cell walls against appressorial penetration. Remove infected crop debris by deep ploughing after harvest. Treat seed with Carbendazim 2 g/kg or Tricyclazole 0.1% solution. PR-121, PR-126                  
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CAUSE	Tungro is caused by a unique complex of two unrelated viruses: Rice Tungro Bacilliform Virus (RTBV, a pararetrovirus) and Rice Tungro Spherical Virus (RTSV, an RNA virus). Both must be present for full symptom expression, though RTSV alone causes mild or no symptoms. The viruses are transmitted in a semi-persistent manner exclusively by the green leafhopper <i>Nephotettix virescens</i>. The leafhopper acquires the virus within minutes of feeding on an infected plant and transmits it to healthy plants almost immediately. There is no transovarial (egg) transmission. The virus does not multiply in the insect vector. RTBV and RTSV are transmitted by the green leafhopper <i>Nephotettix virescens</i>.
SYMPTOMS	Infected plants are severely stunted — they may reach only one-quarter to one-third the height of healthy plants. Leaves show yellow-orange discoloration, typically starting from the tips and spreading towards the base, with a distinctive rusty or chestnut-orange colour on older leaves. Interveinal chlorosis (alternating yellow-green striping) is visible on younger leaves. Tillering is greatly reduced. Flowering is delayed, and panicles that do emerge are poorly filled, small, and have few grains. In severe cases (kresek-like), young seedlings may wilt and die within a week. Infected plants show reduced root development, making them susceptible to lodging. Infected plants are severely stunted and show yellow-orange discoloration on older leaves and interveinal chlorosis on younger leaves. Tillering is reduced and flowering is delayed.
FAVORABLE CONDITIONS	Tungro epidemics are driven by leafhopper population dynamics. Warm, humid weather (25–35°C) promotes rapid leafhopper multiplication. Fields near abundant grassy weed hosts (<i>Echinochloa</i>, <i>Leersia</i>, <i>Oryza sativa</i> f. <i>spontanea</i>) that harbour both the virus and vector are at highest risk. Early transplanted crops and those transplanted close to infected ratoon fields are particularly vulnerable. Regional synchrony failure — when fields at different growth stages are adjacent — allows continuous leafhopper migration between sources and susceptible crops. Warm, humid weather (25–35°C) promotes rapid leafhopper multiplication. Fields near abundant grassy weed hosts are at highest risk.
PREVENTION	Synchronise transplanting in a region — when all fields are transplanted within a 2-week window, the continuous leafhopper bridge is broken. Use tungro-resistant or tolerant varieties (IR-64, TDK-1, PSB-Rc82 have Tungro resistance genes <i>Tvb1</i> and <i>Tsa1</i>). Raise nurseries in insect-proof net houses to ensure virus-free seedlings. Remove and destroy all weed hosts (especially <i>Echinochloa crusgalli</i>, volunteer rice, <i>Leersia hexandra</i>) from field bunds and irrigation channels. Avoid ratoon cultivation in tungro-endemic areas. Do not transplant from infected nurseries — all infected seedlings must be destroyed before they introduce the pathogen. Use tungro-resistant varieties like IR-64.

**TREATMENT &
CONTROL**

Tungro is the most important viral disease of rice in South and Southeast Asia. Yield losses range from 20% in mild outbreaks to near-total crop failure (70–90%) in severe epidemic years. In the 1998–99 Punjab epidemic, thousands of hectares were severely affected. The disease disproportionately affects marginal farmers who cannot afford to replace lost crop income. Chronic low-level infection reduces quality and quantity of rice produced in endemic areas.

IMPACT

SUGARCANE DISEASES



Sugarcane — Yellow Leaf



Sugarcane Yellow Leaf Virus (SCYLV) — Luteovirus

CAUSE

Yellow Leaf is caused by Sugarcane Yellow Leaf Virus (SCYLV), a member of the genus Polerovirus (family Luteoviridae). The virus is phloem-limited, meaning it replicates only in the phloem tissue and cannot be transmitted by mechanical inoculation. Primary vectors are the sugarcane aphid (*Melanaphis sacchari*) and the corn aphid (*Rhopalosiphum maidis*), which transmit the virus in a persistent, circulative manner. The virus can also spread through infected planting setts — making use of infected seed material a major source of introduction into new areas. The virus also infects sorghum, maize, and several wild grasses, which serve as virus reservoirs.

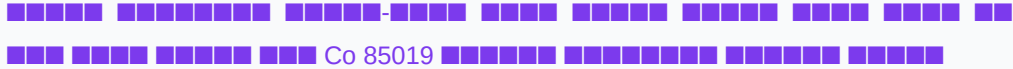
SCYLV (Melanaphis sacchari) (Rhopalosiphum maidis)

SYMPTOMS

The most characteristic symptom is yellowing of the midrib (central vein) of older leaves — the midrib turns bright yellow while the rest of the leaf remains green initially. As disease progresses, yellowing spreads from the midrib to the entire leaf blade, turning leaves pale yellow then straw-coloured, and finally dying from the tip downward. In advanced stages, newly emerging leaves in the spindle also show yellowing, indicating systemic spread. Affected stalks are stunted, produce fewer tillers, and show reduced internode elongation. Root system is weakened, making plants susceptible to lodging and drought stress. Affected canes show reduced brix (sugar content) and produce lower-quality juice.

FAVORABLE CONDITIONS

Hot, dry conditions (35–40°C) favour rapid aphid multiplication and migration, increasing virus spread. Drought stress on plants makes them more susceptible to infection. Densely planted or poorly ventilated fields that create humid micro-environments within the canopy increase aphid persistence. Ratoon crops have higher disease incidence because virus inoculum builds up over multiple seasons. Areas adjacent to reservoirs of alternative hosts (maize, sorghum fields) face higher infection pressure.

PREVENTION	Use only certified, virus-free planting material — preferably tissue-culture plantlets raised from meristems of indexed disease-free plants. Avoid planting setts from fields known to have yellow leaf. Grow tolerant varieties (Co 85019, CoC 671, CoJ 64, CoS 767 show reduced susceptibility). Remove and destroy all infected stools as soon as symptoms are noticed — do not allow ratoon from infected stools. Eradicate weed hosts (Johnson grass, <i>Sorghum halepense</i>, wild maize) from field margins. Avoid growing sugarcane adjacent to maize or sorghum fields where aphid vectors proliferate. 
TREATMENT & CONTROL	There is no curative treatment for virus-infected plants. Control is based on vector management and cultural practices. Aphid vector control: spray Malathion 50% EC at 1–1.5 litre/ha or Dimethoate 30% EC at 1.5 litre/ha at first sign of aphid colonies. Systemic insecticide soil application: Phorate 10G at 6 kg/ha or Carbofuran 3G at 20 kg/ha as soil incorporation at planting suppresses early aphid populations for 6–8 weeks. Imidacloprid 70% WS at 6–8 g/kg seed sett treatment provides good early-season vector protection. Remove and burn all visibly infected stools immediately — they serve as permanent virus reservoirs. Install yellow sticky traps (25 per hectare) to monitor aphid flight and time spray interventions. 
IMPACT	Yellow Leaf causes moderate to significant economic losses — 10–30% reduction in cane yield and 15–25% reduction in sucrose content. Ratoon crops can suffer much higher losses (40–50%) because virus titre increases with successive crops. Affected cane produces poorer juice quality, lower sugar recovery, and the weakened stalks are prone to toppling, making harvesting difficult. The disease is increasingly prevalent as aphid populations grow under climate warming conditions. 

Sugarcane — Red Rot


Glomerella tucumanensis
(Colletotrichum falcatum)

CAUSE	Red Rot is caused by the ascomycete fungus <i>Glomerella tucumanensis</i> (anamorph: <i>Colletotrichum falcatum</i>). It is considered the most destructive sugarcane disease in India. The pathogen is soil- and seed-borne. It survives as sclerotia in soil for 2–3 years and on infected setts and stubble. The fungus penetrates the cane through cut ends of setts during planting, through wounds made by borer insects, or through the node regions during the growing season. Once inside the stalk, it destroys the parenchyma (storage tissue), converting stored sucrose to alcohol and causing the characteristic red discoloration. Glomerella tucumanensis survives as sclerotia in soil for 2–3 years and on infected setts and stubble.
SYMPTOMS	Early external symptoms: leaves on affected stalks turn yellowish then orange-brown from the base upward, usually starting with the outer leaves. Top leaves narrow and roll. Internal stalk symptoms (definitive diagnostic): split the stalk lengthwise — infected pith shows a distinctive red to reddish-brown discoloration with white patches or cross-bands of whitish mycelium. In advanced stages, the internal tissue becomes hollow with a foul alcoholic smell due to sucrose fermentation. Black acervuli (fungal fruiting bodies) appear on the stalk surface. Roots of affected stools rot, causing the entire stool to dry out. In humid conditions, pink sporodochia (masses of conidia) may be visible in the pith. Leaves on affected stalks turn yellowish then orange-brown from the base upward, usually starting with the outer leaves. Top leaves narrow and roll. Internal stalk symptoms (definitive diagnostic): split the stalk lengthwise — infected pith shows a distinctive red to reddish-brown discoloration with white patches or cross-bands of whitish mycelium. In advanced stages, the internal tissue becomes hollow with a foul alcoholic smell due to sucrose fermentation. Black acervuli (fungal fruiting bodies) appear on the stalk surface. Roots of affected stools rot, causing the entire stool to dry out. In humid conditions, pink sporodochia (masses of conidia) may be visible in the pith.
FAVORABLE CONDITIONS	Warm and humid weather (28–32°C, RH 80–90%) during the growing season favours rapid disease spread. Waterlogging and excessive irrigation create anaerobic conditions that weaken root function and promote fungal spread. Susceptible varieties (CoC-24, CoJ-64 in older planting) are at highest risk. Insect borer damage (top borer, internode borer) creates entry wounds that the pathogen exploits. Fields with high organic matter but poor drainage are predisposed. 28–32°C, RH 80–90% during the growing season favours rapid disease spread. Waterlogging and excessive irrigation create anaerobic conditions that weaken root function and promote fungal spread. Susceptible varieties (CoC-24, CoJ-64 in older planting) are at highest risk. Insect borer damage (top borer, internode borer) creates entry wounds that the pathogen exploits. Fields with high organic matter but poor drainage are predisposed.
PREVENTION	Use disease-free, certified seed setts from healthy fields only — avoid planting from fields with known red rot history. Treat setts before planting: hot water treatment at 50°C for 2 hours kills the pathogen in setts without damaging germination. Chemical sett treatment: dip setts in Carbendazim 50% WP at 1 g/litre for 15–20 minutes, or <i>Trichoderma viride</i> suspension at 10 g/litre for 30 minutes. Rotate with non-host crops (paddy, wheat, legumes) for at least 2 seasons in infected fields — do not plant sugarcane consecutively. Control borer insects (top borer, internode borer) with appropriate insecticides to prevent entry wounds. Ensure good field drainage; avoid waterlogging at any stage. Use disease-free, certified seed setts from healthy fields only — avoid planting from fields with known red rot history. Treat setts before planting: hot water treatment at 50°C for 2 hours kills the pathogen in setts without damaging germination. Chemical sett treatment: dip setts in Carbendazim 50% WP at 1 g/litre for 15–20 minutes, or <i>Trichoderma viride</i> suspension at 10 g/litre for 30 minutes. Rotate with non-host crops (paddy, wheat, legumes) for at least 2 seasons in infected fields — do not plant sugarcane consecutively. Control borer insects (top borer, internode borer) with appropriate insecticides to prevent entry wounds. Ensure good field drainage; avoid waterlogging at any stage.

TREATMENT & CONTROL

Remove and burn infected stools immediately on detection — do not leave in field as they spread spores. Apply Carbendazim 50% WP as soil drench at 1 g/litre around affected stools to limit soil spread. Spray Propiconazole 25% EC at 1 ml/litre on remaining plants as a protective measure. For ratoon crops showing infection, it is more economical to destroy the entire affected patch and re-plant with healthy setts after soil treatment with Carbendazim + lime (2:1 ratio). Biological control: *Trichoderma viride* application at 2.5 kg/ha with 50 kg FYM per acre as soil incorporation reduces soil inoculum effectively. Post-infection, deep ploughing followed by lime application (500 kg/ha) reduces soil-borne inoculum.

[illegible]

IMPACT

Red Rot is the most destructive sugarcane disease in India, capable of killing entire stools and shattering crop stands. In severely infected fields, 30–80% of stools can be killed. In Punjab surveys, 5–15% field incidence has been reported in susceptible varieties. Beyond yield loss, red rot destroys sucrose in infected stalks — juice quality drops sharply, causing financial penalties from mills. The pathogen can persist in soil for 2–3 years, forcing long crop rotations.

30–80%

Sugarcane — Rust



Puccinia erianthi (Sugarcane Brown Rust)

CAUSE

Sugarcane Rust is caused by *Puccinia erianthi*, a macrocyclic rust fungus. Like all *Puccinia* species, it produces urediniospores (summer spores) that are responsible for rapid epidemic spread. The fungus is an obligate biotroph — it can only grow and reproduce on living sugarcane tissue. Urediniospores are produced in large numbers in reddish-brown pustules (uredinia) on leaves and are readily dispersed by wind over long distances. No alternate host is known for sugarcane rust, unlike many other *Puccinia* species. The pathogen survives the off-season on ratoon crops and volunteer plants.

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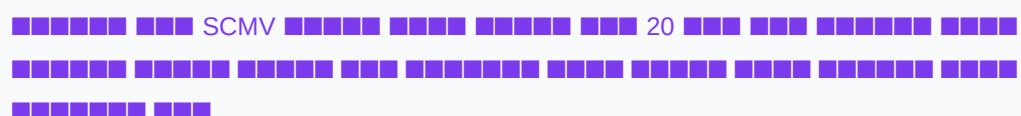
SYMPTOMS	Initial stage: small, pale yellow spots (chlorotic flecks) appear on both surfaces of the leaf blade. Developing stage: spots elongate to form oblong, orange-yellow to reddish-brown pustules (uredinia) — these are visible as raised blisters rupturing through the leaf surface. Each pustule is surrounded by a yellow halo. Advanced stage: pustules turn dark brown to black as teliospores (overwintering spores) replace urediniospores. The leaf tissue around pustules turns necrotic. In severe infections, multiple pustules coalesce, and large portions of leaves turn brown-black and die, reducing the photosynthetic area significantly. Heavily affected leaves look 'rusty' — hence the common name. SYMPTOM INITIAL STAGE DEVELOPING STAGE ADVANCED STAGE HEAVILY AFFECTED LEAF  (1) (2) (3) (4)
FAVORABLE CONDITIONS	Rust develops fastest at moderate temperatures (20–28°C) with high humidity and frequent light rains or heavy dew. Dry conditions after heavy N fertiliser or mill mud (press mud) application create susceptible, succulent tissue ideal for rust infection. Dense crop canopies with poor air circulation maintain the leaf wetness required for spore germination. Young crops (3–5 months) are most susceptible; older crops develop some tolerance. Ratoon crops with continuous green canopy from the previous season allow early rust establishment. 20–28°C HIGH HUMIDITY FREQUENT LIGHT RAINS OR HEAVY DEW DRY CONDITIONS AFTER HEAVY N FERTILISER OR MILL MUD APPLICATION DENSE CROP CANOPIES WITH POOR AIR CIRCULATION YOUNG CROPS (3–5 MONTHS) OLDER CROPS DEVELOP SOME TOLERANCE RATOON CROPS WITH CONTINUOUS GREEN CANOPY FROM THE PREVIOUS SEASON
PREVENTION	Grow rust-resistant or tolerant varieties — Co 99004, CoSe 91010, CoS 8432, Co 87025 show good field resistance. Avoid excessive nitrogen and mill mud application which produces lush, rust-susceptible tissue. Maintain optimum planting density; avoid overcrowding that reduces air circulation within the canopy. Destroy ratoon crops from heavily infected fields to eliminate initial inoculum for the next season. Monitor fields regularly from 3 months after planting when susceptibility is highest. Co 99004 CoSe 91010 CoS 8432 Co 87025
TREATMENT & CONTROL	Apply fungicides at early symptom appearance for best results. Mancozeb 75% WP at 2 kg/ha (2 g/litre) as protective spray — apply 2 sprays at 14-day intervals. Tridemorph 80% EC at 1 litre/ha is a systemic fungicide effective against established rust. Propiconazole 25% EC at 1 ml/litre provides both protective and curative action. Azoxystrobin 23% SC at 0.7 ml/litre combined with Tebuconazole gives excellent rust control with residual protection of 3–4 weeks. Remove and burn heavily infected leaves in small fields before spores spread further. Ensure good drainage to reduce leaf wetness duration. MANCOZEB 75% WP TRIDEMORPH 80% EC PROPICONAZOLE 25% EC AZOXYSTROBIN 23% SC TEBUCONAZOLE

Rust is generally considered a moderate disease, but significant epidemics do occur. Yield losses of 5–15% are typical, but severe outbreaks can cause up to 30% reduction in cane yield and 20% reduction in sucrose content. Photosynthetic area loss is the primary mechanism — plants with rust-damaged leaves cannot sustain normal sucrose accumulation. In new rust-susceptible variety introductions, initial epidemics can be particularly severe before farmers adjust variety choice.



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Sugarcane Mosaic is caused primarily by Sugarcane Mosaic Virus (SCMV), a member of the genus Potyvirus, along with Sorghum Mosaic Virus (SrMV) and occasional Sugarcane streak mosaic virus. The virus infects sugarcane, maize, sorghum, and many grass species. Transmission occurs in a non-persistent manner by at least 20 aphid species (*Melanaphis sacchari*, *Rhopalosiphum maidis*, *Aphis gossypii*). Non-persistent transmission means aphids acquire the virus within seconds of probing and transmit it almost immediately to new plants without requiring a prolonged feeding period. Infected planting setts are the major means of long-distance spread into new regions.



The classic symptom is a mosaic or mottle pattern of alternating light green and dark green patches on leaves — this gives the disease its name. On young leaves, the mosaic pattern is most striking — irregular light green, yellow-green, or yellow patches alternate with normal dark green areas in a 'puzzle-like' pattern. Older leaves may show fading of the colour contrast but still exhibit a streaked or chlorotic appearance. Infected plants show moderate stunting and reduced tillering. Stalks are thinner with shortened internodes. In severe infections, leaves may show necrotic (dead) streaks and premature death of older leaves. No gummosis or oozing is associated with mosaic.



Mosaic incidence is highest in dry periods when aphid populations migrate and move between plants in search of food. Proximity to maize or sorghum fields — major alternative hosts — greatly increases infection pressure. Early-planted crops are more vulnerable as the young, tender tissue is more susceptible to aphid probing and virus inoculation. Fields with high weed pressure provide aphid refuges and reservoirs.



PREVENTION

Use certified virus-free planting material — do not plant setts from mosaic-affected fields. Grow resistant or tolerant varieties where available. Eradicate alternative hosts: remove all Johnson grass (*Sorghum halepense*), volunteer maize, and weedy sorghum from field margins and irrigation channels. Do not intercrop sugarcane with maize, sorghum, or other grass crops as this amplifies aphid vector populations. Rogue out and destroy mosaic-infected plants as soon as spotted — the sooner infected plants are removed, the lower the spread to neighbours.

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TREATMENT & CONTROL

No curative treatment is available for virus-infected plants — management relies entirely on prevention and vector control. Remove and burn all visibly infected stools and their stumps to reduce inoculum. Control aphid vectors: Malathion 50% EC at 1 litre/ha or Thiamethoxam 25% WG at 100 g/ha applied at aphid colony detection. Mineral oil sprays (1–2%) applied to young crop disrupt non-persistent aphid transmission by interfering with stylet activity. Yellow water traps (30 per hectare) can monitor aphid flight to time interventions. Do not ratooning from mosaic-infected stools as virus builds up through successive ratoon crops.

$\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(x) e^{-x^2} dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(x) e^{-(x-i0)^2} dx$ (1)

IMPACT

Mosaic causes relatively minor losses in well-managed fields — typically 5–20% yield reduction in mildly infected crops. However, in regions where virus-free planting material is not used consistently, chronic infection levels can be high with cumulative losses of 20–30%. Juice quality (brix and sucrose content) is reduced in heavily infected canes. The disease is particularly challenging because of the large number of alternative hosts and the difficulty in completely eliminating aphid populations.

[illegible]

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Puccinia triticina (*P. recondita*
f.sp. *tritici*)

CAUSE

Puccinia tritici

SYMPTOMS

2020 年 12 月 31 日 2020 年 12 月 31 日
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FAVORABLE CONDITIONS

[illegible]

PREVENTION

Grow varieties with effective Lr-gene resistance — in Punjab, HD-3086, GW-322, WH-1105 carry multiple Lr genes. Timely sowing (November 1–15 in Punjab) ensures the crop passes through vulnerable flowering and grain fill before peak rust pressure in March–April. Delayed sowing extends the period when crops are in susceptible stages under rising temperatures. Avoid excessive nitrogen application, particularly late top-dressing, which produces susceptible tissue. Destroy volunteer wheat near fields immediately after harvest to eliminate overwintering inoculum. Monitor fields from February using regular scouting for early detection.

[illegible]

TREATMENT & CONTROL

Spray at first detection — do not wait for disease to spread: Propiconazole 25% EC at 1 ml/litre is the recommended systemic fungicide, effective against brown rust. Tebuconazole 25.9% EW at 1 ml/litre provides longer residual protection (3–4 weeks). Mancozeb 75% WP at 2.5 g/litre is a protective spray and should be applied preventively during high-risk periods. Sulfur 80% WP at 3 g/litre is an economical option for early-stage rust. Two sprays at 14-day intervals are typically sufficient. Spray timing is critical: spraying at flag leaf or ear emergence gives maximum protection to the most photosynthetically important leaves.

[illegible]

IMPACT

Brown Rust is endemic to wheat-growing areas of India and occurs every year to some degree. In normal years, losses are 5–15% on susceptible varieties. In epidemic years with extended warm, humid spells in March–April, losses of 20–30% have been recorded in Punjab and Haryana. Early infection (before heading) is more damaging than late infection. The economic threshold for fungicide application is generally 5–10% infected leaf area.

[illegible]

Wheat — Yellow Rust



Puccinia striiformis f.sp. *tritici*

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TREATMENT & CONTROL

Act at first sign of infection — even a single pustule per field signals an urgent need to spray: Propiconazole 25% EC at 1 ml/litre — highly effective, rapid curative action, repeat at 14-day intervals. Tebuconazole 25.9% EW at 1 ml/litre gives 3–4 weeks residual protection. Combination products: Propiconazole + Trifloxystrobin (Nativo 75 WG at 0.5 g/litre) provide both protectant and curative activity with extended residual. Hexaconazole 5% EC at 2 ml/litre is a cost-effective option. Early spray (before flag leaf stage) is far more effective than late spray. Sulfur 80% WP at 3 g/litre is protectant only — not effective on established infections.

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IMPACT

Yellow Rust has the highest epidemic potential of all wheat rusts. On susceptible varieties, 40–70% yield loss is possible in severe epidemic years, with some reports of near-complete crop failure in unprotected fields. In a bad year (e.g. 2008–09 in Punjab and Haryana), losses exceeded Rs 3,000 crore in India. The emergence of new 'super races' (e.g. Warrior, Triticale Aggressive) that overcome previously resistant varieties has made yellow rust management increasingly challenging. Early detection and prompt fungicide application are the only reliable protection when resistant varieties are not available.

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Wheat — Septoria Leaf Blotch



Parastagonospora nodorum /
Zymoseptoria tritici

CAUSE

Septoria Leaf Blotch is caused by two distinct but related pathogens: *Zymoseptoria tritici* (formerly *Mycosphaerella graminicola*) causing Septoria Tritici Blotch (STB) — the more globally important species, and *Parastagonospora nodorum* (formerly *Septoria nodorum*) causing Septoria Nodorum Blotch (SNB). *Z. tritici* produces pycnidia (asexual fruiting bodies) that exude conidia in humid conditions. *P. nodorum* produces both pycnidia (on leaves and glumes) and ascospores (sexual stage). Both pathogens survive on infected crop residue and stubble and can be seed-borne. *P. nodorum* in particular causes severe glume blotch and grain infection.

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SYMPTOMS	Septoria Triticum Blotch (Z. tritici): pale green to yellow, irregular blotches that turn tan-brown with age. The definitive diagnostic feature is the presence of tiny black pycnidia (fruiting bodies) visible as black dots within the lesion — best seen with a hand lens. Lesions are bounded by leaf veins, giving them a rectangular or angular shape. Yellow chlorosis surrounds lesions. Septoria Nodorum Blotch (P. nodorum): oval to elongated tan lesions on leaves, but critically also infects glumes — brown-black lesions on glumes ('glume blotch') that cause floret sterility and grain shrivelling. In severe infections, entire leaves die from the tip downward. Heavily infected canopies appear blighted from a distance. Septoria Nodorum (P. nodorum) causes oval to elongated tan lesions on leaves, but critically also infects glumes — brown-black lesions on glumes ('glume blotch') that cause floret sterility and grain shrivelling. In severe infections, entire leaves die from the tip downward. Heavily infected canopies appear blighted from a distance.
FAVORABLE CONDITIONS	Both pathogens require cool, wet conditions — temperatures of 10–25°C with prolonged leaf wetness from rain, dew, or irrigation. Rain events are critical for dispersal — pycnidia release conidia in splash droplets during rain, which are distributed upward through the canopy with successive rain events. Dense canopies that maintain high humidity and reduce drying between rains extend infection cycles. Late-sown crops in wet years with frequent rain are most affected. Susceptible varieties under high nitrogen are particularly vulnerable. 10–25°C with prolonged leaf wetness from rain, dew, or irrigation. Rain events are critical for dispersal — pycnidia release conidia in splash droplets during rain, which are distributed upward through the canopy with successive rain events. Dense canopies that maintain high humidity and reduce drying between rains extend infection cycles. Late-sown crops in wet years with frequent rain are most affected. Susceptible varieties under high nitrogen are particularly vulnerable.
PREVENTION	Use certified disease-free seed, treated with Carboxin + Thiram (Vitavax Power) 2–2.5 g/kg to eliminate seed-borne infection. Grow varieties with partial resistance — no fully immune varieties are commercially available, but several have adult plant resistance genes. Incorporate crop residue by deep ploughing after harvest to accelerate decomposition of infected straw (the primary overwintering reservoir). Rotate out of wheat for at least 2 seasons in fields with high Septoria pressure. Avoid overhead irrigation during cool, cloudy weather that extends leaf wetness. Reduce planting density slightly in fields with previous Septoria history. Use certified disease-free seed, treated with Carboxin + Thiram (Vitavax Power) 2–2.5 g/kg to eliminate seed-borne infection. Grow varieties with partial resistance — no fully immune varieties are commercially available, but several have adult plant resistance genes. Incorporate crop residue by deep ploughing after harvest to accelerate decomposition of infected straw (the primary overwintering reservoir). Rotate out of wheat for at least 2 seasons in fields with high Septoria pressure. Avoid overhead irrigation during cool, cloudy weather that extends leaf wetness. Reduce planting density slightly in fields with previous Septoria history.
TREATMENT & CONTROL	Apply fungicide at tillering (GS 30–32) in high-risk situations, especially in wet seasons: Mancozeb 75% WP at 2.5 g/litre for 2 sprays 14 days apart as a protectant. Propiconazole 25% EC at 1 ml/litre provides both protectant and curative action, applied at flag leaf emergence. Combination fungicides: Propiconazole + Cyproconazole or Azoxystrobin + Cyproconazole — excellent residual control for 3–4 weeks. Trifloxystrobin + Tebuconazole (Nativo) at 0.5 g/litre is highly effective against both STB and SNB. Application at flag leaf (GS 37) and ear emergence (GS 59) protects the most important yield-determining growth stages. Apply fungicide at tillering (GS 30–32) in high-risk situations, especially in wet seasons: Mancozeb 75% WP at 2.5 g/litre for 2 sprays 14 days apart as a protectant. Propiconazole 25% EC at 1 ml/litre provides both protectant and curative action, applied at flag leaf emergence. Combination fungicides: Propiconazole + Cyproconazole or Azoxystrobin + Cyproconazole — excellent residual control for 3–4 weeks. Trifloxystrobin + Tebuconazole (Nativo) at 0.5 g/litre is highly effective against both STB and SNB. Application at flag leaf (GS 37) and ear emergence (GS 59) protects the most important yield-determining growth stages.

IMPACT

Septoria diseases are the most economically important foliar diseases of wheat in Europe and are becoming increasingly prevalent in South Asia under changing rainfall patterns. Yield losses in wet years typically range from 5–20% but can reach 30–40% in severe epidemics on susceptible varieties. *P. nodorum* additionally causes significant grain quality problems through glume blotch — infected grain has higher mycotoxin levels, reduced protein content, and lower test weight. The increasing frequency of wet seasons associated with climate change may make Septoria more important in Punjab in coming decades.

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Wheat — Loose Smut



Ustilago tritici

CAUSE

Loose Smut is caused by *Ustilago tritici*, a basidiomycete fungus. It is a unique seed-borne systemic pathogen — the fungus infects the wheat flower during the current season and lies dormant as mycelium within the embryo of the seed. When infected seed is planted, the fungus grows systemically with the plant through all growth stages, completely replacing the grain tissue with black teliospores at heading. There are no external symptoms until heading — the infection is entirely internal and invisible. Seeds from smutted plants look normal but harbour dormant mycelium, making visual selection of seed impossible without laboratory testing.

Ustilago tritici is a seed-borne systemic pathogen that infects wheat flowers during the current season and lies dormant as mycelium within the embryo of the seed. When infected seed is planted, the fungus grows systemically with the plant through all growth stages, completely replacing the grain tissue with black teliospores at heading. There are no external symptoms until heading — the infection is entirely internal and invisible. Seeds from smutted plants look normal but harbour dormant mycelium, making visual selection of seed impossible without laboratory testing.

SYMPTOMS

No symptoms are visible at any growth stage until the plant heads. At heading (ear emergence): smutted heads emerge slightly earlier than healthy ones. Instead of normal grain-bearing spikelets, all florets of the entire head are replaced by a mass of olive-brown to black powdery teliospores enclosed in a thin membrane. The membrane soon ruptures in the wind, releasing a cloud of black smut powder and leaving only the naked rachis (central spike axis). After spore dispersal, the rachis looks like a bare, twisted spine — completely bare of any grain. Adjacent healthy plants may become infected at heading via airborne spores landing on open flowers — infection is entirely through the flower. Plants infected at heading this season will not show symptoms until the following crop season.

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